

MID TERM TEST 2024/2025

Duration: 1 hour 30 minutes

Instructions:

1) You can only refer to lecture notes. \ 2) Save this file as your MatricNo-MTT. \ 3) File should be saved as .ipynb file. \ 4) Submission will be done after the test ended. \ 5) Please upload the file in the designated folder in the e-learning. \ 6) The access to the e-learning will only be closed 10 minutes after the test ended.

Please run the below code to import Numpy and Pandas

In []:

```
import numpy as np
import pandas as pd
```

QUESTION 1 [40 Marks]

Use Python Basics to answer this questions.

✓ Question 1 (a) - 10 marks

Write a program that calculates the Body Mass Index (BMI) and categorizes the result into 4 different weight statuses. \ The formula for BMI is $\text{weight} / \text{height}^2$, where weight is in kilograms and height is in meters.

Categories:

Underweight: $\text{BMI} < 18.5$

Normal weight: $18.5 \leq \text{BMI} < 24.9$

Overweight: $25 \leq \text{BMI} < 29.9$

Obesity: $\text{BMI} \geq 30$

In [1]:

```
#Write your code here
```

✓ Question 1 (b)- 10 marks

A university scores student matriculation number in the following format:

MAT20231234

Here :

- " MAT " is a prefix that represent faculty of Mathematics
- The next four digits represent the year of enrollment

- The last four digits are the unique student ID

Question 1(b)(i) - 3 marks \ Extract the year of enrollment form a given matriculation number

In [3]: `#Write your code here`

Question 1(b)(ii) - 3 marks \ Check if a given matriculation number starts with the prefix " MAT "

In [3]: `#Write your code here`

Question 1(b)(iii) - 4 marks \ Write a function " generate_matric_number(fac, year, student_id) " that returns a valid matriculation number in the same format when provided with a year of enrollment and a unique student ID and for faculty of Mathematic.

In [3]: `#Write your code here`

Question 1(c)- 10 marks

You are provided with a text names story contain the following content

"Once upon a time, in a land far away, there lived a young programmer who loved to solve problems. He spent his days learning Python and exploring the secrets of code."

Story="Once upon a time, in a land far away, there lived a young programmer who loved to solve problems. He spent his days learning Python and exploring the secrets of code."

Question 1(c)(i)- 1 Mark \ Find the length of the string and print it

In [3]: `#Write your code here`

Question 1(c)(ii)- 2 Marks \ Identify and print the first occurrence of the word " Python " in the text using an appropriate string method

In [6]: `#Write your code here`

Question 1(c)(iii)- 2 Marks \ Extract the substring " upon a time " using string slicing and print it

In [7]: `#Write your code here`

Question 1(c)(iv)- 3 Marks \ Write a code that to display only " there lived a young programmer who loved to solve problems "

In [8]: `#Write your code here`

Question 1(c)(v)- 2 Marks \ Use string slicing to reverse the text and pring the reversed string

In [9]: `#Write your code here`

Question 1(d)- 3 Marks

Define a lambda function that takes three numbers as inputs and returns their average.\nDemonstrate the function by calculating the average of 10, 20, and 30.

In [9]: `#Write your code here`

In [10]: `#Write your code here`

Question 1(e)- 5 Marks

Given the following following tuple of tuples:

```
data = ((1, "Alice", 25), (2, "Bob", 30), (3, "Charlie", 35))
```

In [10]: `#Write your code here`

Question 1(e)(i) - 2 marks\ Extract and print the tuple (3, "Charlie", 35) from data.

In [10]: `#Write your code here`

Question 1(e)(ii) - 1 mark\ Calculate and print the length of the tuple data.

In [10]: `#Write your code here`

Question 1(e)(iii) - 2 marks\ Unpack the tuple (2, "Bob", 30) into three variables: id, name, and age , and print these variables.

In [10]: `#Write your code here`

Question 1(f) - 2 marks

Given these two list

```
list1 = [1, 2, 3, 4]
list2 = [5, 6, 7, 8]
```

Question 1(f)(i)-1 mark\ Concatenate list1 and list2 into a single list and print the resulting list

In [10]: `#Write your code here`

Question 1(f)(ii)- 1 marks\ Add the element 9 to the concatenated list and print the updated list.

```
In [ ]: #Write your code here
```

QUESTION 2 [20 MARKS]

Use NumPy to answer the following questions.

Question 2(a)(i) - 2 marks Create a NumPy array with 30 random integers between 1 and 100.

```
In [ ]: #Write your code here
```

Question 2(a)(ii) - 2 marks Reshape the NumPy array into a 5x6 matrix

```
In [15]: #Write your code here
```

Question 2(a)(iii) - 7 marks

(1) Extract all elements from the third column (Col3) - 2 marks (2) Compute the sum of the elements in the fourth column (Col4) - 2 marks (3) Create a new column with the element-wise sum of Col3 and Col4 - 2 marks (4) Display the results - 1 marks

```
In [ ]: #Write your code here
```

Question 2(b)(i) - 2 marks

Create a 2D NumPy array with dimensions 5x4, filled with random integers between 10 and 50.

```
In [ ]: #Write your code here
```

Question 2(b)(ii) - 2 marks

Sort each row in ascending order.

```
In [ ]: #Write your code here
```

Question 2(b)(iii) - 5 marks

Find the sum of each column and the maximum value in the entire array.

```
In [ ]: #Write your code here
```

QUESTION 3 [34 MARKS]

Question 3(a) i - 3 marks

Create a Pandas DataFrame with the following data. Save this DataFrame to a CSV file named sales.csv.

TransactionID	Date	Region	Product	Salesperson	Units Sold	Unit Price	Total Sales
1001	1/1/2024	North	Laptop	Alice	5	1200	6000
1002	1/2/2024	South	Tablet	Bob	3	800	2400
1003	1/3/2024	East	Smartphone	Charlie	10	700	7000
1004	1/4/2024	West	Laptop	David	2	1500	3000
1005	1/5/2024	North	Tablet	Eve	7	900	6300

In []: *#Write your code here*

Question 3(a) ii - 2 marks

Load the `sales.csv` file into a new Pandas DataFrame and display its content.

In []: *#Write your code here*

Question 3(b) - 5 marks

(i) Display the initial rows of the datasets. (1 mark)\ (ii) Provide an overview of the columns, data types, and counts of non-null values. (2 marks)\ (iii) Check if the data has missing values. (2 marks)

In []: *#Write your code here*

Question 3(c) i - 2 marks

Add a new column Profit to the DataFrame, calculated as 20% of the Total Sales column.

In []: *#Write your code here*

Question 3(c) ii - 4 marks

Create a column Total Revenue by multiplying Units Sold and Unit Price (verify it matches Total Sales).

In []: *#Write your code here*

Question 3(c) iii - 3 marks

Filter out the transactions where Units Sold is less than 5 and display the result.

In []: *#Write your code here*

Question 3(c) iv - 3 marks

Rename the column Salesperson to Employee and save the updated DataFrame to a new CSV file named `sales_updated.csv`.

In []:

```
#Write your code here
```

Question 3(d) i - 4 marks

Group the data by Region and calculate the total units sold in each region.

In []:

```
#Write your code here
```

Question 3(d) ii - 3 marks

Find the average Total Sales for each product type.

In []:

```
#Write your code here
```

Question 3(d) iii - 5 marks

Find the salesperson who sold the maximum number of units and display their detail

In []:

```
#Write your code here
```